Carlos Andrés Montoya Tobón



Senior Java Backend Engineer & Solutions Architect

Medellín, Colombia · andresmontoyat@gmail.com · +57 324 442 2196 · github.com/andresmontoyat · linkedin.com/in/carlos-andres-montoya-tobon · andresmontoyat.co

CORE STACK Java · Spring Boot · Microservices · AWS · Hexagonal Architecture · Keycloak · PostgreSQL · RabbitMQ

PROFESSIONAL SUMMARY

I'm a Java Solution Architect and Tech Lead based in Medellín, Colombia, with 18+ years in backend engineering: 11+ of them leading teams of up to 45 people, and nearly 7 as a solutions architect. I've designed and built Java platforms ranging from transportation management systems and telecom self-service portals to fintech payments, digital identity and blockchain-verified ticketing. My specialty is backend architecture on the Spring ecosystem: hexagonal microservices and event-driven systems on Kafka and RabbitMQ, built for resilience and idempotency, secured with Keycloak and observable end to end, deployed on AWS, GCP and Azure. As a lead I set the architecture, grow the team technically and own delivery, backed by DevOps culture and agile practice (certified Scrum Master).

EXPERIENCE

Solution Architect & Tech Lead

May 2026 — Present

Soldife · Remote

- Set the platform's architectural standard — hexagonal + DDD across 6 microservices and 9 SDKs, dependencies always pointing to the domain, enforced by ArchUnit on every build (zero architecture drift between PRs).
- Designed an ecosystem of 9 auto-configurable Spring Boot SDKs with contract governance — SemVer, backward compatibility and ADRs — published versioned to GitHub Packages.
- Established cross-cutting patterns: transactional outbox, canonical event envelope, end-to-end W3C tracing and centralized identity at the gateway.
- Documented architecture as code: C4 model (L1–L4) in Structurizr, ADRs and AI-oriented development guidelines.
- Designed CI/CD on GitHub Actions (GitFlow + shift-left) with reusable workflows parameterized by project type (Java, KrakenD, NGINX, contracts) and mandatory quality gates: build → tag-filtered unit/Karate tests → JaCoCo coverage → OWASP scan → ArchUnit → Spotless.
- Built SemVer-tagged base images on GHCR (Temurin 21, GraalVM native-image, KrakenD/Go toolchain) to cut per-job tool installs and speed pipelines; org-level governance via branch/PR rulesets and in-pipeline Slack notifications.

Java · Spring Boot · Hexagonal Architecture · DDD · Microservices · ArchUnit · GitHub Actions · KrakenD · GraalVM · Structurizr

Backend Developer

Jan 2026 — Apr 2026

CODERIO · Remote

- Developed backend REST APIs for ITTI, a Paraguayan bank, on Java, Kotlin and Spring Boot behind a Kong API gateway.
- Improved the performance of the core Personas (customers) API — cut response times and raised throughput, profiled and monitored with New Relic.

Java · Kotlin · Spring Boot · Kong · New Relic · REST APIs · Microservices

Backend Developer

Sep 2024 — Jan 2026

F2X - Flypass · Medellín, Colombia

- Designed a commissions-calculation component for the F2X memberships product (Java · Spring Boot · hexagonal architecture on AWS · Postgres).
- Designed a billing component for the F2X fuel-services product across the microservice fleet.
- Owned production-incident response and operational support for backend services.

Java · Spring Boot · Microservices · Hexagonal Architecture · AWS · PostgreSQL

Backend Developer

Dec 2023 — Dec 2024

Blerify · Remote

- Built security components with Keycloak and implemented KrakenD as the platform's API gateway.
- Developed a blockchain-based voucher/bond issuance component (Java · Vert.x · Kotlin · blockchain · Postgres on GCP/Kubernetes).

Java · Vert.x · Kotlin · Keycloak · KrakenD · Blockchain · PostgreSQL · GCP · Kubernetes

Backend Developer

Apr 2022 — Jun 2023

Tul · Medellín, Colombia

- Built warehouse-management modules for the TUL B2B marketplace microservice fleet (Java · Spring Boot · hexagonal architecture on AWS · Postgres).
- Delivered Spring Security components and ran platform-wide code-quality reviews.

Java · Spring Boot · Spring Security · Microservices · Hexagonal Architecture · AWS · PostgreSQL

Solutions Architect
Aug 2020 — Apr 2022

k4s - Klever for Solutions · Medellín, Colombia

- Architected PaaS solutions on AWS for MutualSER's next-gen health-insurance products (Colombian EPS).
- Designed reusable security and Keycloak/GCP integration frameworks consumed across product teams.
- Mentored developers on Spring, SOLID principles and clean architecture.

Java · Spring Boot · AWS · Keycloak · GCP

Solutions Architect
May 2016 — Apr 2020

Integral Software SAS · Medellín, Colombia

- Solutions Architect designing and evolving backend architectures on the Spring ecosystem over a four-year engagement.
- Designed multiple architectures and systems for the Mayor's Office of Medellín — among them Atenea and Tienda Virtual — applying hexagonal architecture, design patterns and security.
- Built components to migrate legacy systems to new architectures for Universidad de Medellín.
- Delivered the same legacy-to-modern architecture migration for the Colombian Air Force (FAC).

Kotlin · Java · Spring Boot · PostgreSQL · React · Angular · Keycloak · Hexagonal

Solutions Architect
Sep 2015 — May 2016

Software Estratégico · Medellín, Colombia

- Defined the new architectures for Matrix Tech's product line on Spring Boot, RabbitMQ and Cassandra.
- Built an SMS messaging gateway from scratch.
- Led the adoption of a DevOps culture to improve communication and collaboration across IT areas, enabling faster, more efficient delivery.

Java · Spring Boot · RabbitMQ · Cassandra · DevOps

Technical Lead
Aug 2013 — Sep 2015

Tata Consultancy Services · Medellín, Colombia

- Led a distributed team of 45+ developers building Claro Perú's transactional self-service portal, from design through to its production release.
- Designed and built the Oracle Service Bus integration layer connecting Claro Colombia's back-office systems.
- Built the portal itself on Java 8, Spring and WebSphere Portlets, front end included, against IBM WebSphere.
- Stood up the continuous-integration infrastructure and introduced Scrum, giving a 45-person team a single repeatable delivery cadence.

Java · Spring · WebSphere Portlets · Oracle Service Bus · Jenkins

Lead Developer
Nov 2010 — Aug 2013

PATH S.A. · Medellín, Colombia

- Designed NetPlanet analysis components for the Claro network RCs and vehicle-tracking platforms.
- Built Mail Bomber and SMS-delivery systems in Java; IoT pilots for the city of Medellín.
- Designed AWS and on-premise infrastructure and CI/CD pipelines across the organization.

Java · AWS · IoT · CI/CD

Java Web Developer
Jun 2009 — Jun 2010

Ceiba Software House · Medellín, Colombia

- Built the loan, policy and occupational-risk systems of Suramericana — Colombia's largest insurer — on Java / JEE 5 with Struts and JSF.
- Wrote the Oracle data layer behind those flows: SQL and stored procedures carrying the policy and claims operations.
- Delivered every slice full-stack — Java back end, JavaScript / HTML / CSS front end and Oracle schema — deployed on Tomcat.

Java · JEE 5 · Struts · JSF · Oracle SQL · Tomcat

Java Web Developer
Apr 2007 — Jun 2009

Castor SAS (antes Mercurio) · Medellín, Colombia

- Designed and built the business components of a Transportation Management System that ran the daily operation of 25+ transport companies.
- Built the integration layer that carried TMS operations into each client's accounting system, removing the manual re-keying between both worlds.
- Owned support and maintenance for every company on the platform — production incidents, SQL Server / MySQL data fixes and JSP + JavaScript front-end work.
- Introduced Scrum and extreme-programming practices to a team that until then delivered ad hoc.

Java · Servlets · JSP · JavaScript · SQL Server · MySQL · Tomcat

LIVE SITES

MyPipelines **mypipelines.andresmontoyat.co** — Reusable CI/CD pipeline library — parameterized GitHub Actions workflows (build, tag-filtered tests, JaCoCo coverage, OWASP scan, ArchUnit, Spotless) shared across JVM, KrakenD and NGINX projects.

MyGuidelines — Engineering Standards **myguidelines.andresmontoyat.co** — My engineering playbook, codified — the architecture, testing, security and clean-code standards I hold every project to, and the trade-offs behind them. Hexagonal + DDD for JVM microservices, written to be executable by engineers and AI coding agents.

TECHNICAL SKILLS

Backend	**Java** · **Kotlin** · Spring Boot · Spring Cloud · Spring Security · Hibernate · Kafka · RabbitMQ · Cassandra · REST / gRPC · Keycloak
Cloud & Infrastructure	**AWS** · GCP · Azure · **Kubernetes** · Docker · GKE · KrakenD · Terraform
DevOps & Tools	Jenkins · GitHub Actions · SonarQube · Nexus · Gradle · **Maven** · PostgreSQL · MySQL · Oracle SQL · Scrum Master
AI & Productivity	**Claude Code** · GitHub Copilot · JetBrains Junie · Prompt Engineering